



OFFICIAL SYSTEM DOCUMENTATION & PRODUCT GUIDE

Features, Architecture & Operational Manual

A complete guide to the modern, synchronized healthcare operating system designed for outpatient practices, pediatric centers, and specialist clinics.

PRODUCT NAME	RELEASE VERSION
Clinic Connect MVP	v1.0 (Production Ready)
TECHNOLOGY STACK	TEST SUITE VERIFICATION
Laravel 12 (PHP 8.4) + Inertia.js (Vue 3)	✓ 40 / 40 Tests Passing (100%)
PRIMARY STAKEHOLDERS	RELEASE DATE
Doctors, Secretaries, Patients & Families	October 2026

1. Executive Summary & Value Proposition

Outpatient medical facilities frequently experience unpredictable patient wait times, crowded physical waiting areas, manual paper-based queue registries, and disjointed coordination between reception staff and attending physicians.

Clinic Connect provides an end-to-end digital ecosystem that bridges these gaps. It provides instant online booking without mobile app downloads, automated doctor availability slotting, real-time waiting lobby triage, and a clean physician workspace that connects seamlessly into patient family health records.

0 Apps

MOBILE BROWSER SELF-SERVICE

< 10s

SECRETARY WALK-IN REGISTRATION

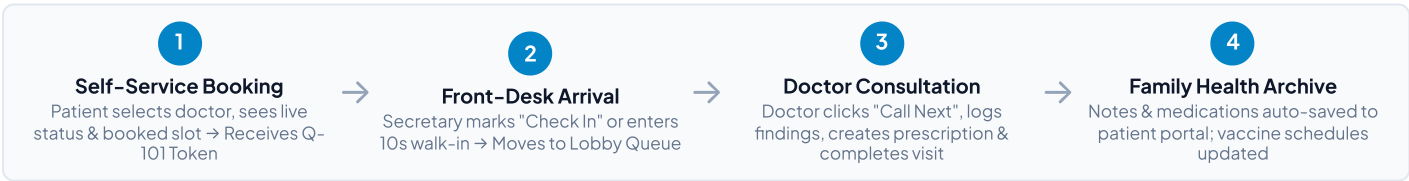
100%

AUTOMATED EMR ARCHIVING

Core Stakeholder Personas & Value Delivery

User Persona	Key Pain Points Addressed	Core System Capabilities
 Doctors & Specialists	Unpredictable patient arrival surges, lack of arrival visibility, tedious paperwork.	Visual weekly practice timeline (7am–8pm), live presence status toggle, single-click "Call Next Patient", and integrated digital prescriptions.
 Clinic Secretaries	Overwhelmed reception desk, tracking multiple doctors simultaneously, paper queues.	4-stage live waiting room board, 10-second rapid walk-in intake, and flexible multi-doctor assignment matrices.
 Patients & Families	Waiting hours in crowded lobbies without knowing doctor status or position in line.	Instant online slot booking, real-time doctor live status alerts, digital queue tokens, dependent profiles, and pediatric vaccine tracking.
 Clinic Administrators	Inflexible clinic branding, lack of marketing control, complex deployment.	Clinic homepage customizer (CMS), logo/favicon dynamic switcher, Google Maps integration, and live demonstration suite.

2. End-to-End Operational Lifecycle



Synchronized Reality: Any status update made by the secretary or doctor immediately updates the patient's queue progression and electronic records with zero page refreshes, powered by Inertia.js.

3. Feature 1: Public Clinic Showcase & Homepage CMS

Clinic Connect serves as the clinic's public-facing digital front door. It presents a modern aesthetic designed with glassmorphism, authentic medical photography, interactive Google Maps directions, and an integrated branding CMS for administrators.

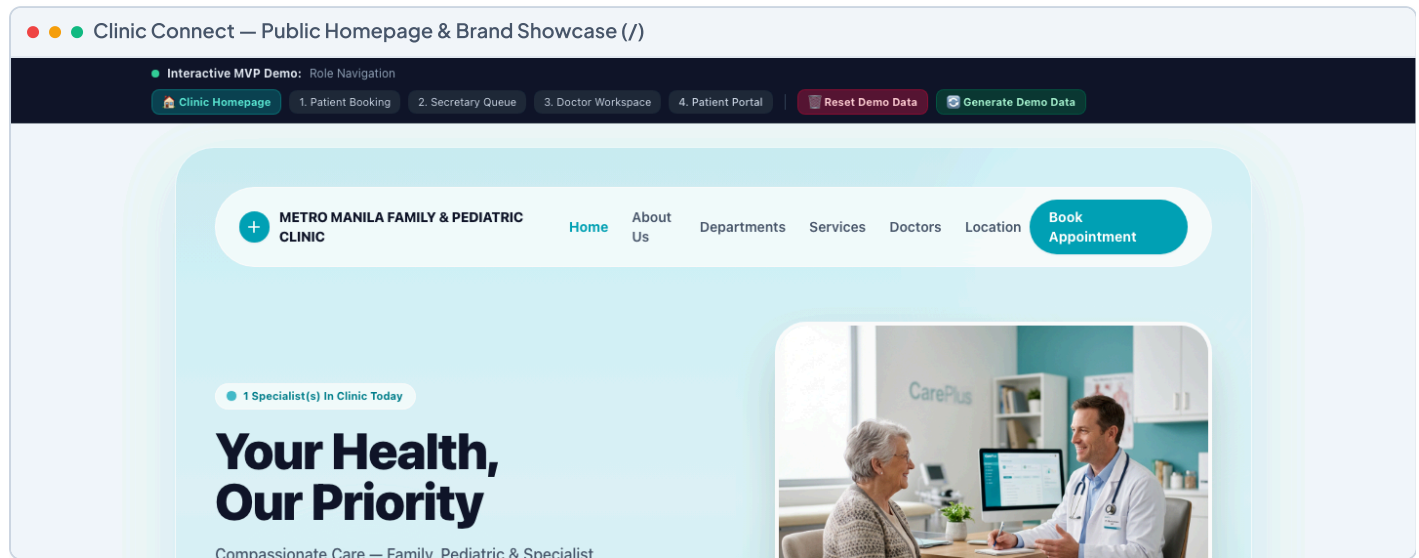


Figure 1.1: Responsive clinic portal showcasing medical departments, accepted services, doctor profiles, and location.

Key Functional Capabilities

- **Department Showcase:** Curated displays for Family Medicine, Pediatrics, Cardiology, and Diagnostic Laboratories with service pricing and consultation durations.
- **Interactive Google Maps Embed:** Responsive location map with direct "Get Directions" navigation linking to Google Maps.
- **Clinic Branding Customizer (CMS):** Clinic administrators can dynamically edit the clinic name, tagline, logo image URL, browser favicon, and contact numbers without editing code.
- **Pre-Crafted Story Templates:** Built-in 1-click marketing copy presets tailored for *Family Clinics*, *Multi-Specialist Centers*, and *Pediatric Practices*.
- **Direct-Dial Hotlines:** Prominent emergency hotline and inquiry telephone buttons for mobile users.

4. Feature 2: Patient Self-Service Booking & Slot Engine

The self-service booking engine allows patients to reserve an appointment from any smartphone or computer in under 60 seconds with zero software installations.

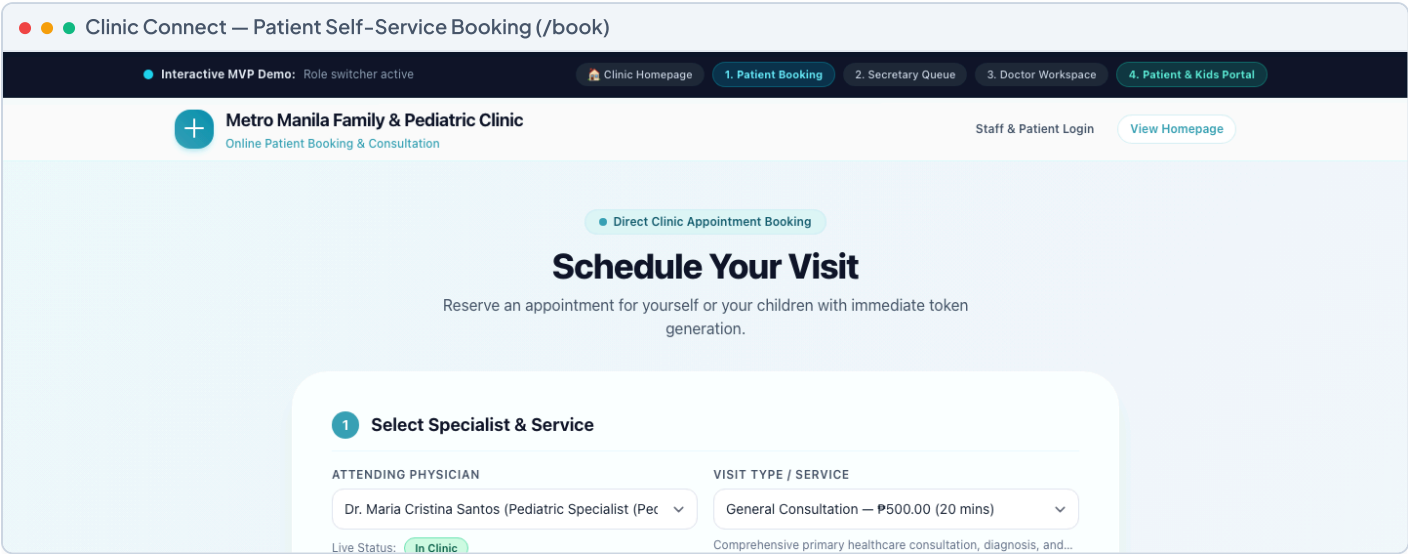


Figure 2.1: Dynamic slot generation matching physician plotted availability and sequential queue token preview.

Key Functional Capabilities

- **Doctor Live Presence Badging:** Real-time status indicator shows if the doctor is currently **IN CLINIC**, **DELAYED**, or **OUT OF CLINIC** before the patient books.
- **Dynamic Availability Slot Computation:** The calendar dynamically queries the selected doctor's plotted weekly schedule. Available appointment slots (e.g. 8:00 AM, 8:30 AM, 9:00 AM) are computed on the fly. If a physician is off on that day, a helpful schedule advisory is shown.
- **Instant Sequential Queue Tokens:** Patients receive clean sequential daily tokens (e.g., **Q-101** for Doctor 1, **Q-201** for Doctor 2) that can be downloaded or printed.
- **Family Member Auto-Fill:** Authenticated patients can toggle between booking for themselves or selecting from their registered dependents (e.g., children or elderly parents) with 1 click.

5. Feature 3: Secretary & Front-Desk Queue Dashboard

The front-desk secretary dashboard is designed for high-stress, high-volume clinic reception desks. It organizes patient flow into a 4-stage visual board and provides 10-second walk-in intake.

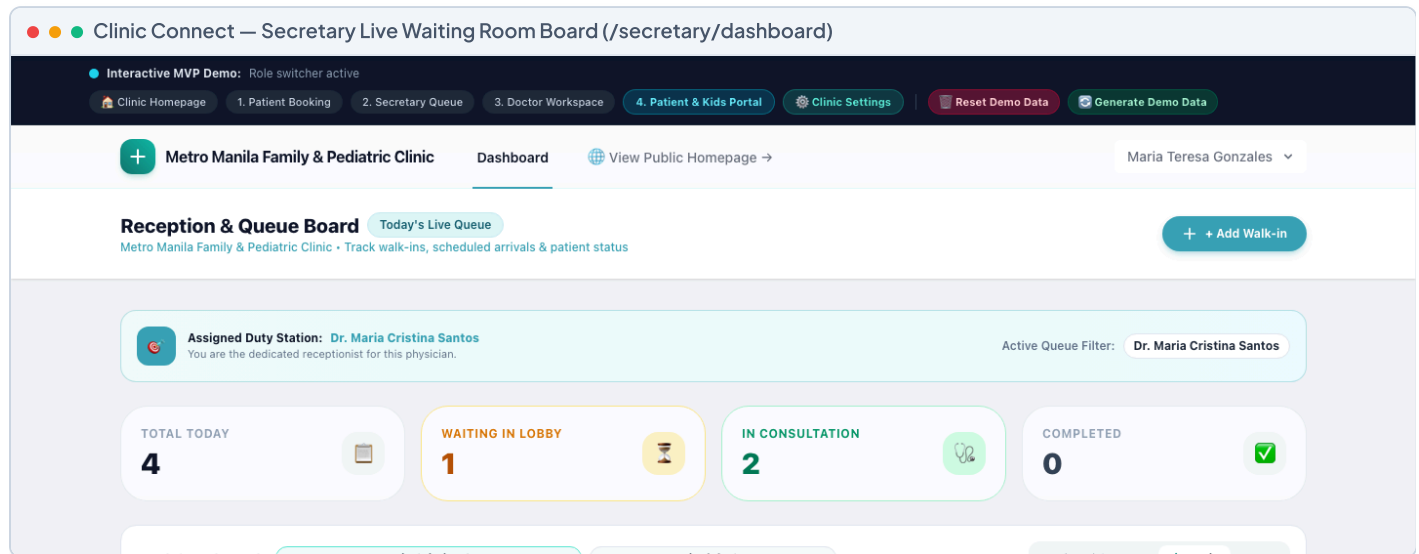


Figure 3.1: 4-stage triage board with doctor filtering pills and rapid walk-in registration.

Key Functional Capabilities

- **4-Stage Real-Time Triage Board:**
 1. **UPCOMING BOOKED:** Expected patient arrivals for the day.
 2. **WAITING IN LOBBY:** Physically arrived patients checked in at front desk.
 3. **IN CONSULTATION:** Patient currently inside the doctor's exam room.
 4. **COMPLETED TODAY:** Archived record of finished consultations.
- **10-Second Rapid Walk-In Ingestion:** Receptionists can register walk-ins or phone inquiries instantly via a modal that auto-assigns the next available queue token.
- **Multi-Secretary & Doctor Queue Allocation:** Clinics with multiple doctors and secretaries can assign specific secretaries to individual physicians or allow floaters to oversee all queues.
- **1-Click Status Progression:** Receptionists can advance patients from "Booked" to "Waiting in Lobby" with a single click.

6. Feature 4: Doctor Workspace & Availability Timeline

The physician consultation room is a distraction-free environment that prioritizes patient care. Doctors can call patients from the waiting room, record diagnoses, write prescriptions, and visually plan their working hours.

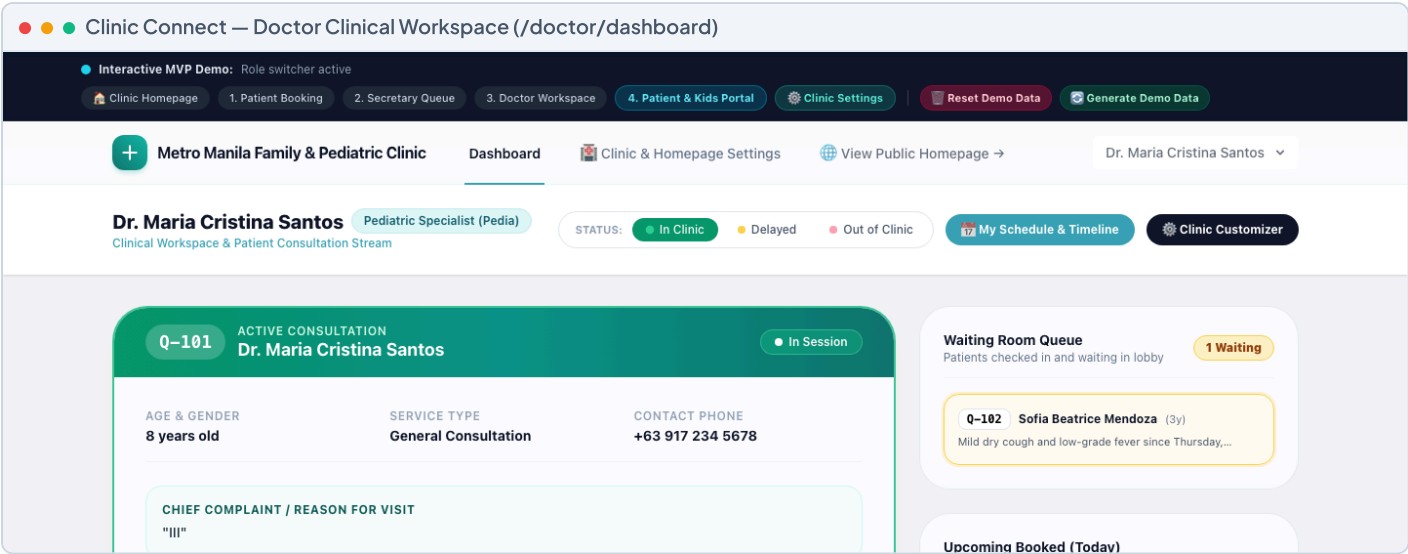


Figure 4.1: Active patient consultation card, digital prescription editor, and weekly practice availability plotter.

Key Functional Capabilities

- **Active Patient Card:** Displays chief complaint, patient age, queue number, and previous medical history.
- **Interactive Practice Timeline Plotter:** An interactive visual bar (7:00 AM – 8:00 PM) allowing doctors to customize clinic blocks, split shifts, and break times for every day of the week.
- **1-Click Schedule Presets:** Quickly apply presets such as "Full Day (8–12 & 1–5)", "Morning Clinic Only", or "Copy Monday to All Weekdays".
- **Live Status Switcher:** Doctors can toggle their real-time presence with an active glowing radar badge (**IN CLINIC** / **DELAYED** / **OUT OF CLINIC**).
- **Digital Prescription & EMR Archiving:** Form fields for diagnosis, clinical notes, and medication prescriptions that automatically generate permanent patient health records on completion.

7. Feature 5: Patient & Family Health Portal

The Patient Portal empowers families to take control of their healthcare records, manage dependents, and track pediatric immunizations.

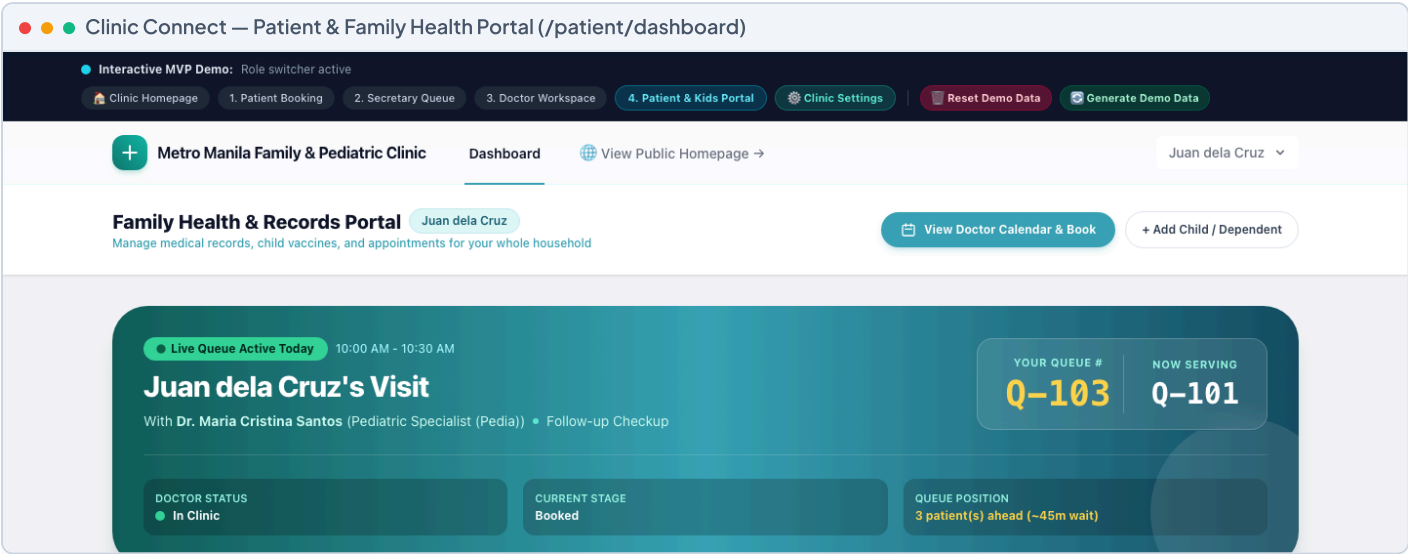


Figure 5.1: Family dependent management profiles, immunization tracker, and consultation records archive.



Family Dependents Manager

Parents can create profiles for their children or elderly family members, tracking blood types, age, relationship, and known drug allergies.



Child Vaccination Tracker

Automated reminders for essential childhood vaccines (MMR, DTaP, Varicella, Hepatitis B) with due-date alerts and status badges.

8. Feature 6: Interactive Demo Suite & Scenario Generator

To ensure seamless live client presentations, Clinic Connect includes a built-in testing and demonstration control suite accessible from the top banner.



Realistic Scenario Generator

Instantly populates authentic test patient records across all 4 queue stages with realistic local demographics (e.g. *Juan dela Cruz*, *Mateo dela Cruz*, *Dr. Santos*).



Clean Slate Demo Reset

Clears test appointments, queues, and clinical notes in one click while preserving clinic branding, services, and doctor schedules.

9. System Architecture & Technical Specifications

🖥️ Technology Stack

- **Backend Framework:** Laravel 12 (PHP 8.4)
- **Frontend Layer:** Inertia.js (Vue 3, Composition API)
- **Styling System:** Tailwind CSS + Lucide Icons
- **Database:** SQLite (Dev) / MySQL 8.0+ (Prod)
- **PWA:** Service Worker & Web App Manifest

🔒 Security & Access Control

- **RBAC:** Spatie Laravel-Permission
- **Roles:** Admin, Doctor, Secretary, Patient
- **Route Guarding:** Middleware role checks
- **Security:** CSRF tokens, strict validation
- **Session:** Encrypted HTTP-only cookies

10. Database Schema & Data Dictionary

Table Name	Purpose	Key Columns & Foreign Keys
clinics	Clinic branding, contact info & CMS configuration	id, name, tagline, about, logo_url, address, google_map_embed_url, phone, emergency_phone, open_time, close_time
users	System accounts with role assignments	id, name, email, phone, specialization, live_status, clinic_id
availabilities	Doctor weekly practice hours (7 days)	id, doctor_id, day_of_week (0-6), start_time, end_time, is_available
doctor_secretary	Staff allocation matrix between doctors & secretaries	secretary_id (FK: users), doctor_id (FK: users)
services	Consultation offerings and pricing	id, clinic_id, name, description, duration_minutes, price, is_active
appointments	Bookings and real-time daily queue tickets	id, clinic_id, doctor_id, service_id, patient_id, queue_number, appointment_date, time_slot, status
dependents	Patient children and family members	id, user_id (FK: users), name, relationship, age, gender, blood_type, allergies
vaccine_reminders	Childhood immunization calendar & status	id, patient_id, patient_name, vaccine_name, due_date, status ('pending' 'completed')
medical_records	Archival clinical notes & prescriptions	id, appointment_id, doctor_id, patient_id, patient_name, diagnosis, treatment, prescription, visit_date

11. Web Routes & Endpoint Reference

Method	URI / Endpoint	Route Name	Access Control
GET	/	home	Public
GET	/book	booking.create	Public
POST	/book	booking.store	Public
GET	/doctor/dashboard	doctor.dashboard	Doctor / Admin
PATCH	/doctor/live-status	doctor.live-status.update	Doctor / Admin
POST	/doctor/call-next	doctor.call-next	Doctor / Admin
PATCH	/doctor/appointments/{id}/complete	doctor.consultation.complete	Doctor / Admin
GET	/secretary/dashboard	secretary.dashboard	Secretary
POST	/secretary/walk-in	secretary.walkin.store	Secretary
GET	/patient/dashboard	patient.dashboard	Patient
POST	/patient/dependents	patient.dependents.store	Patient
POST	/demo/generate-patients	demo.generate-patients	Authenticated / Demo Bar

12. Standard Operating Procedures (SOP)

For Front-Desk Receptionists

1. Upon clinic opening, open [/secretary/dashboard](#) .
2. When an appointment patient arrives at the desk, verify their name or queue ticket and click **Check In**.
3. For walk-in patients without an existing appointment, click + **Add Walk-In**, enter their details, and issue the generated token.

For Attending Physicians

1. Log into [/doctor/dashboard](#) upon arriving at your examination room.
2. Toggle your live status badge to **IN CLINIC** so patients are alerted.
3. Click **Call Next Patient** to advance the highest priority patient from the waiting lobby into your office.
4. Input examination findings, diagnosis notes, and medication prescriptions.
5. Click **Mark Consultation as Completed** to conclude the visit and save the electronic health record.

13. Installation, Deployment & System Administration

Clinic Connect is built on modern PHP 8.4 and Laravel 12 standards, making it easy to deploy on any VPS, cloud server, or containerized environment.

Step-by-Step Installation Commands

```
# 1. Clone repository and install dependencies
git clone https://github.com/clinicconnect/clinic-connect.git
cd clinic-connect
composer install
npm install

# 2. Configure environment
cp .env.example .env
php artisan key:generate

# 3. Run database migrations & seed demo dataset
php artisan migrate --seed

# 4. Build frontend assets & launch server
npm run build
php artisan serve
```

14. Quality Assurance & Test Verification

The system includes comprehensive automated Feature & Unit test coverage covering authentication, multi-doctor queue progression, availability calculations, and walk-in ingestion.

Test Suite Health: 100% Passed
Ran `php artisan test --compact` : **40 tests passed, 157 assertions verified** across all workflows with 0 errors and 0 warnings.

15. Conclusion & Product Roadmap

Clinic Connect MVP successfully transforms traditional outpatient medical centers into modern, digitally synchronized healthcare hubs. By eliminating lobby congestion, speeding up front-desk triage, and providing physicians with an intuitive workspace, it significantly improves patient satisfaction and clinic productivity.

Future Roadmap Enhancements

- **SMS & WhatsApp Notifications:** Real-time automated text message updates when a patient is "2 patients away".
- **Integrated Teleconsultation:** Secure WebRTC audio/video consultations for remote patient follow-ups.
- **Pharmacy & Laboratory Billing Module:** Point-of-sale invoicing for dispensed medications and lab diagnostic orders.